The Hopf Graph as a Lattice Gauge Structure: A Framework Observation*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*

(Dated: April 15, 2026)

The GeoVac framework places Coulomb bound-state quantum numbers on a discrete graph that converges to the unit three-sphere S^3 in the continuum limit. This paper documents a framework observation: the same graph, viewed through its incidence structure, instantiates the combinatorial ingredients of a lattice gauge theory. Nodes carry scalar matter (0-forms), edges carry a $U(1)$ connection (1-forms), and the Hopf quotient $S^3 \to S^2$ (collapsing the magnetic quantum number m_l along its $2l+1$ fiber) produces the discrete base graph on which gauge-neutral propagation lives. The node Laplacian $L_0 = BB^\top$ is the matter propagator of the graph; the edge Laplacian $L_1 = B^\top B$ is the discrete Hodge–1 Laplacian, whose kernel counts independent cycles and whose structure is the combinatorial analog of a photon propagator in temporal gauge. Nothing about the mathematical decomposition is new: graph Hodge theory (Lim; Schaub *et al.*), lattice gauge theory (Wilson), and the conformal invariance of 4D Maxwell (Cunningham; Bateman) are all established separately. What is new is the synthesis: the GeoVac graph is a concrete case where all three structures coexist on a single, physically motivated discretization. Under the proposed interpretation of Paper 2, in which the bundle $S^1 \to S^3 \to S^2$ is read as (gauge phase)–(electron states)–(photon momenta), the combinatorial formula $K = \pi(B + F - \Delta)$ becomes a cross-sector spectral invariant of this lattice gauge structure. We do not claim a new derivation of the fine-structure constant. We do claim that the Paper 2 Observation belongs inside an established mathematical frame—one that makes its components (selection principle, combination rule, circulant Hermiticity) into gauge-theoretic statements rather than free-floating numerology. We close with open questions: whether the S^5 Bargmann–Segal lattice of the 3D harmonic oscillator admits an analogous SU(3) gauge structure, and whether the rank-2 Breit extension of Paper 22 lifts the $U(1)$ 1-form picture to a higher-form gauge theory.

I. INTRODUCTION

The computational framework documented in Papers 0, 7, 14, 16, 18, 22, and 24 reformulates non-relativistic Coulomb bound-state physics as spectral graph theory on a discretization of the unit S^3 [17, 19–21, 23**?** , 24]. The graph's nodes are quantum-number triples (n, l, m_l), its edges are the $L_\pm$ and $T_\pm$ ladder-operator transitions between neighboring states, and its graph Laplacian converges to the Laplace–Beltrami operator on the unit S^3 in the continuum limit (Paper 7, eighteen symbolic proofs).

Separately, Paper 2 proposes that the Hopf fibration $S^1 \to S^3 \to S^2$—the same S^3 that hosts the electron's bound-state manifold—is more than a geometric coincidence. In the Paper 2 reading, S^3 is the electron state space, S^2 is the photon momentum sphere, and S^1 is the $U(1)$ gauge phase [18]. The fine-structure constant is then reinterpreted as the cost of projecting S^3 onto S^2, i.e. the information absorbed in the fiber. A combinatorial formula

$$K = \pi\,(B + F - \Delta) \tag{1}$$

reproduces α^{-1} at 8.8×10^{-8} with zero free parameters and p-value 5.2×10^{-9} (Paper 2 [18]). Seven sprints of structural decomposition (Phases 4B–4I) have identified each of B, F, Δ as spectrally natural objects on distinct corners of the Hopf bundle, but the additive combination rule itself remains unexplained.

This paper makes one observation. The GeoVac Hopf graph is simultaneously:

1. *A discretization of* S^3 (Fock projection; Paper 7).

2. *A simplicial complex carrying a discrete Hodge decomposition* (Lim [12]; Schaub *et al.* [13]).

3. *A lattice gauge structure* in the Wilson sense [2], with matter on nodes and a $U(1)$ connection implicit on edges through the ladder-operator phases.

No single paper in the literature has articulated (1)–(3) as one object. Lattice gauge theory has been formulated on flat hypercubic lattices and, separately, on topologically nontrivial manifolds (S^4 for instanton physics [3, 4], $S^3 \times \mathbb{R}$ for finite-volume confinement studies [5]). Graph Hodge theory has been applied to signal processing and network analysis [12, 13]. Conformal Maxwell theory on the conformal boundary of AdS_5 [7, 10] is a separate fourth thread. The communities have not overlapped on the specific case where all four threads meet. The closest published precedent for items (1)–(3) collected on a single graph object is the "gauge networks" framework of Marcolli and van Suijlekom [14], which defines systems of finite spectral triples on graphs whose spectral

* Paper 25 in the Geometric Vacuum series

action is a Wilson lattice gauge theory; the present GeoVac construction is a specific instance of that framework at the data level (with the continuum limit of the original gauge networks, as corrected by the Perez-Sanchez Comment [16], being Yang–Mills without Higgs — the quiver extension [15] obtains Yang–Mills–Higgs for its own enlarged construction); the forced-graph claim (Bertrand $\times$ Fock projection) and the master Mellin engine are GeoVac-original structural extensions not predicted by the MvS scaffolding.

We do not claim a new derivation of α. We do claim that the Paper 2 observation and its proposed interpretation can be placed inside an established mathematical framework. In that framework, B is a Casimir-weighted node-trace on the S^2 quotient graph; F is a spectral zeta of the infinite Fock-degeneracy lattice at the packing exponent; Δ is a boundary-mode count on the squared-Dirac sector. The $\mathbb{Z}_3$ circulant structure of the cubic self-consistency equation (Paper 2 [18]) becomes a threefold projection structure on the three bundle components. What remains open is not the existence of these quantities—they are computable and have been computed in Phases 4B–4G [18]—but their additive combination (an Observation, not a derivation; §13.5 standing label) and its proposed physical interpretation.

Scope. This is an observational synthesis paper. No new experimental or computational prediction is derived. The purpose is (i) to document a framework connection that has not been made in either the GeoVac papers or the broader literature, (ii) to place the open components of Paper 2 (the observed combination rule and its proposed interpretation) inside an established mathematical vocabulary, and (iii) to identify open questions (e.g., the S^5 Bargmann–Segal analog, the Breit tensor extension) that become natural within the lattice-gauge framing. The paper is a companion to Paper 21's broader synthesis [22], focused on one specific structural observation.

What is open and what is sharp. The *mathematical* decomposition of the Hopf graph into node-Laplacian, edge-Laplacian, and S^2-quotient components is sharp: these are exactly computable, exactly described in Sections II–III. The *physical* interpretation of the bundle components (electron/photon/gauge) is Paper 2's proposed interpretation, flagged throughout as such. The interpretive gap is deliberate: we do not assert the interpretation is correct. We assert that if it is correct, the mathematics already in the literature places it in a specific lattice-gauge context.

The remainder of this paper is organized as follows. Section II surveys graph Hodge theory in the form we need (incidence matrix, node/edge Laplacians, discrete Hodge decomposition, Betti numbers). Section III describes the GeoVac Hopf graph and its S^2 quotient in this language. Section IV states the framework observation formally and traces Paper 2's (B, F, Δ) ingredients to corresponding spectral invariants. Section V discusses what the observation does and does not predict. Section VI reviews related work and discusses why the gap remained. Section VII lists open questions. Section VIII concludes.

II. GRAPH HODGE THEORY: THE TOOLS

This section collects the combinatorial machinery we use in Sections III–IV. The content is standard; our contribution is the identification in Section III that the GeoVac graph is exactly the kind of simplicial 1-complex to which this machinery applies.

A. Incidence matrix and boundary operators

Let $G = (V, E)$ be an undirected graph with $|V| = N_0$ nodes and $|E| = N_1$ edges. Fix an orientation on each edge (a choice, but the Laplacians below are orientation-independent). The signed node-edge incidence matrix $B \in \mathbb{R}^{N_0 \times N_1}$ has entries

$$B_{iv,\,e} = \begin{cases} +1 & \text{if edge } e \text{ starts at node } v, \\ -1 & \text{if edge } e \text{ ends at node } v, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Geometrically, B is the discrete exterior derivative from 0-cochains (scalar functions on nodes) to 1-cochains (scalar functions on oriented edges), $d_0 = B^\top$, and its transpose B is the discrete boundary operator ∂_1 from 1-chains to 0-chains.

B. Node and edge Laplacians

The combinatorial Hodge–0 and Hodge–1 Laplacians are [12, 13]

$$L_0 = BB^\top \in \mathbb{R}^{N_0 \times N_0}, \tag{3}$$

$$L_1 = B^\top B \in \mathbb{R}^{N_1 \times N_1}. \tag{4}$$

L_0 is the familiar node Laplacian $D - A$ (degree matrix minus adjacency matrix). L_1 is the edge Laplacian, acting on functions defined on edges. If the simplicial complex has triangular 2-faces with incidence matrix B_2, the full Hodge–1 Laplacian acquires an upper piece $B_2 B_2^\top$; we work with triangle-free simplicial 1-complexes throughout this paper, so $L_1 = B^\top B$ is the complete edge Laplacian.

C. Discrete Hodge decomposition

The cochain space $\mathbb{R}^{N_1}$ (edge functions) decomposes orthogonally:

$$\mathbb{R}^{N_1} = \mathrm{im}(d_0) \oplus \ker(L_1). \tag{5}$$

The first summand contains *gradient* edge functions, $\omega = d_0 f = B^\top f$ for some node function f—the edge function is "pure gauge" in the sense that it is a gradient. The second summand, $\ker(L_1)$, contains the *harmonic* 1-cochains: edge functions that are not gradients but whose Hodge–1 Laplacian vanishes. By the combinatorial Hodge theorem, the dimension of the harmonic subspace equals the first Betti number of the graph:

$$\dim\ker(L_1) = \beta_1 = N_1 - N_0 + c, \tag{6}$$

where c is the number of connected components. β_1 counts independent cycles: edge functions that circulate without being gradients. These are the combinatorial analog of a non-trivial holonomy along a loop, i.e., the analog of a Wilson loop in the Wilson formulation of lattice gauge theory.

On a manifold, the full Hodge–1 Laplacian is $\Delta_1 = d_0 d_0^* + d_1^* d_1$, where the first term is the *exact* (longitudinal) part and the second is the *co-exact* (transverse) part. On a bare graph (simplicial 1-complex without 2-cells), only the exact part $L_1 = B^\top B = d_0 d_0^*$ is available. The co-exact part requires faces (2-cells) to define a boundary operator d_1; Paper 30 [27] provides precisely these faces via plaquette enumeration, giving the co-exact Laplacian $L_1^{\text{co}} = d_1 d_1^\top$. Paper 28 [26] shows that the transverse propagator $(d_1 d_1^\top)^+$ recovers the continuum self-energy structural zero that L_1^+ alone cannot, establishing a concrete QED role for the face structure.

D. Node versus edge spectra

The nonzero eigenvalues of $L_0 = BB^\top$ and $L_1 = B^\top B$ coincide by the singular-value decomposition: if $B = U\Sigma V^\top$ then $L_0 = U\Sigma^2 U^\top$ and $L_1 = V\Sigma^2 V^\top$. The edge Laplacian therefore contributes no new eigenvalues, only new kernel dimensions (the β_1 harmonic 1-forms beyond the c harmonic 0-forms). This relation, elementary in matrix algebra, will matter in Section IV: the edge Laplacian's *topological* content (counting cycles) is distinct from its *spectral* content (shared with the node Laplacian).

E. Lattice gauge theory dictionary

The ingredients of Wilson's [2] lattice gauge theory map directly to the graph-Hodge ingredients:

Wilson	Graph Hodge	Role
Site x	Node v	Matter location
Link $(x, x+\hat{\mu})$	Edge e	Gauge-field location
Link variable $U_e \in G$	Edge phase $\theta_e \in [0, 2\pi)$	$U(1)$ connection
Plaquette P	Triangle/cycle	Curvature / $F_{\mu\nu}$
Wilson loop W_C	Holonomy $\prod_{e\in C} e^{i\theta_e}$	Gauge-invariant observable
Plaquette action $\mathrm{Re}\,\mathrm{tr}\,U_P$	Curvature norm $\lvert F\rvert^2$	Gauge kinetic term
Gauge transformation	$\theta_e \to \theta_e + (\phi_{t(e)} - \phi_{h(e)})$	0-coboundary shift
Matter propagator	$(L_0 + m^2)^{-1}$	Covariant Laplacian
Photon propagator	$(L_1 + m^2)^{-1}$	Hodge–1 resolvent

The match is exact at the level of combinatorial data. The only question is whether a given graph carries the physical content that makes this dictionary nontrivial. For a random graph with arbitrary $U(1)$ phases, the dictionary is empty: there is no physical interpretation of the gauge degrees of freedom. For the GeoVac Hopf graph, the ladder-operator structure provides a canonical assignment of edge phases through the angular-momentum and radial transition amplitudes. Section III makes this explicit.

III. THE GEOVAC HOPF GRAPH

A. Nodes and edges

The GeoVac graph at cutoff $n_{\max}$ has nodes labeled by (n, l, m_l) with $n = 1, \ldots, n_{\max}$, $l = 0, \ldots, n-1$, and $m_l = -l, \ldots, +l$. The total node count is $N_0(n_{\max}) = \sum_{n=1}^{n_{\max}} n^2$. Edges come in two kinds [19, 20]:

- **Angular** edges connect $(n, l, m_l) \leftrightarrow (n, l, m_l \pm 1)$ at fixed (n, l)—these are the $L_\pm$ transitions of the orbital angular-momentum ladder.
- **Radial** edges connect $(n, l, m_l) \leftrightarrow (n', l, m_l)$ where $n' = n \pm 1$ (and sometimes, depending on the discretization scheme, with $\Delta l = \pm 1$)—these are the $T_\pm$ hydrogenic radial transitions.

At $n_{\max} = 3$ the graph has $N_0 = 1 + 4 + 9 = 14$ nodes and $N_1 = 13$ edges, with $c = 3$ connected components (s-chain, p-block, d-chain). The first Betti number is $\beta_1 = N_1 - N_0 + c = 13 - 14 + 3 = 2$ [25]. The two independent cycles both live in the p-block, where the angular m_l-ladder and the radial n-transition together form square-like loops.

B. The Hopf quotient: $S^3 \to S^2$

The Hopf fibration $S^1 \to S^3 \to S^2$ projects the S^3 onto the base $S^2 \cong \mathbb{CP}^1$, collapsing each S^1 fiber to a point. On the GeoVac graph, the natural discrete implementation of this quotient collapses each magnetic fiber $\{(n, l, m_l) : m_l = -l, \ldots, +l\}$ at fixed (n, l) to a single sector label (n, l). The Hopf base is then a graph on (n, l) sectors.

At $n_{\max} = 3$, the base graph has 6 sectors,

$$\{(1,0), (2,0), (2,1), (3,0), (3,1), (3,2)\}, \tag{7}$$

with sector weights $(2l+1) = 1, 1, 3, 1, 3, 5$ (the fiber dimensions). Inter-sector edges of the base graph count the number of S^3 edges crossing between sectors; intra-sector angular edges collapse into "fiber" structure that does not appear in the base. Track α-D [18] computed this base graph explicitly: its Laplacian has eigenvalues $\{0, 0, 0, 1, 3, 6\}$ (three zero modes reflecting the three disconnected components inherited from the S^3 graph). The quotient construction and the exact spectrum are pinned live in `tests/test_paper25_s2_quotient.py` (sympy-exact: characteristic polynomial $x^3(x-1)(x-3)(x-6)$; crossing-count weights $1, 1, 3$ on the s-chain and p-block radial bundles, d-chain sector isolated).

C. $U(1)$ phase on edges

The $L_\pm$ and $T_\pm$ ladder operators act on the spherical harmonic basis with computable matrix elements. For angular edges,

$$\langle n, l, m_l + 1 | L_+ | n, l, m_l \rangle = \hbar \sqrt{l(l+1) - m_l(m_l+1)}\, e^{i\phi_{L_+}}. \tag{8}$$

For radial edges, the amplitudes are the hydrogenic Clebsch–Gordan coefficients times a phase. Under a global $U(1)$ rotation of the wavefunctions $\psi_{n,l,m_l} \to e^{i\chi}\psi_{n,l,m_l}$, the edge amplitudes pick up no phase; under a node-dependent $\psi \to e^{i\chi_v}\psi$, the edge amplitude for $e = (v, v')$ picks up the phase $e^{i(\chi_{v'} - \chi_v)}$—this is exactly the $U(1)$ gauge transformation in the Wilson dictionary of Section II.E.

The GeoVac graph is therefore naturally equipped with a $U(1)$ bundle: each edge carries a complex phase, and the phases transform covariantly under node-local $U(1)$ redefinitions of the wavefunctions. The curvature of this bundle is the holonomy around each closed cycle. At $n_{\max} = 3$, $\beta_1 = 2$ cycles (both in the p-block) carry nontrivial holonomy.

D. Node and edge Laplacians of the GeoVac graph

With the incidence matrix B as in Eq. (2), the node Laplacian $L_0 = BB^\top$ is the one whose continuum limit is $-\Delta_{S^3}$ (Paper 7). The edge Laplacian $L_1 = B^\top B$ has explicit spectrum at $n_{\max} = 3$ computed in Track Q1–B [25]:

$$\mathrm{spec}(L_1) = \{0^{(2)}, \tfrac{3-\sqrt{5}}{2}, 1^{(2)}, \tfrac{5-\sqrt{5}}{2}, 2, \tfrac{3+\sqrt{5}}{2}, 3^{(3)}, \tfrac{5+\sqrt{5}}{2}, 5\}. \tag{9}$$

The two zero modes confirm $\beta_1 = 2$. (The full spectrum is pinned sympy-exactly, live from `GeometricLattice`, in `tests/test_paper25_s2_quotient.py`, 2026-07-04.) The golden-ratio eigenvalues come from the path graph P_5 of the d-chain; the eigenvalues $\{1, 3, 3, 5\}$ come from the p-block (which contains both cycles). The characteristic polynomial factors completely over $\mathbb{Q}(\sqrt{5})$: all edge-Laplacian spectral invariants are algebraic numbers with denominator dividing 2^k times a power of $\sqrt{5}$.

Observation 1 follows directly:

Observation 1 (Edge Laplacian is algebraic on a finite Hopf graph)**.** *The edge Laplacian $L_1 = B^\top B$ of the GeoVac graph at any finite $n_{\max}$ has integer-valued entries, hence algebraic eigenvalues. No transcendental spectral invariant can live on the edge Laplacian of a finite Hopf graph. Transcendentals enter only through spectral zeta functions in the continuum limit ($n_{\max} \to \infty$), consistent with the Phase 4C–4D finding that $F = \pi^2/6$ is not accessible from any finite-graph Hopf-fiber trace [18].*

E. The matter and gauge propagators

Under the Paper 2 interpretation:

- $L_0 = BB^\top$ is the matter propagator's inverse: bound electron states propagate by the S^3 Laplace–Beltrami operator in the continuum.
- $L_1 = B^\top B$ is the gauge-sector propagator's inverse: the $U(1)$ photon-like edge modes, in temporal gauge, propagate by the discrete Hodge–1 Laplacian. The kernel dimension β_1 counts independent Wilson-loop classes.

This is the observational claim of the paper. We now formalize it.

IV. THE FRAMEWORK OBSERVATION, FORMALIZED

Proposition 1 (Lattice gauge structure of the GeoVac graph)**.** *Let G be the GeoVac graph at cutoff $n_{\max}$, with nodes labeled by (n, l, m_l), edges labeled by angular and radial ladder-operator transitions, and signed incidence matrix B. Then:*

1. *G is a triangle-free simplicial 1-complex, so its combinatorial Hodge Laplacians are $L_0 = BB^\top$ and $L_1 = B^\top B$.*
2. *The node Laplacian L_0 converges to the Laplace–Beltrami operator $-\Delta_{S^3}$ in the continuum limit (Paper 7 [19]).*
3. *The edge Laplacian L_1 has kernel dimension $\beta_1(G) = N_1 - N_0 + c$; these harmonic 1-forms are the discrete analog of topologically nontrivial $U(1)$ connections.*
4. *The Hopf quotient (collapsing m_l-fibers) yields a sub-graph G^{S^2} on (n, l) sectors, with Laplacian spectrum $\{0, 0, 0, 1, 3, 6\}$ at $n_{\max} = 3$ (Track α-D [18]; pinned exactly in `tests/test_paper25_s2_quotient.py`).*
5. *Natural $U(1)$ phases on edges arise from the ladder-operator amplitudes; a node-local redefinition $\psi_v \to e^{i\chi_v}\psi_v$ acts as a gauge transformation in the Wilson sense on these phases.*

This proposition collects facts already scattered across Papers 0, 2, 7, 14, and the Phase 4B Track α-D data set. The new content is the unified *reading* of those facts. The reading is not a new computational claim; it is a dictionary that places the GeoVac graph in the Wilson–Hodge frame.

A. Where Paper 2's (B, F, Δ) live in this frame

The three structural ingredients of Paper 2's combination rule $K = \pi(B + F - \Delta)$ have, by the Phase 4B–4H sprint program, been localized on distinct components of the Hopf bundle. In the present framework, the localizations acquire gauge-theoretic labels.

$B = 42$*: Casimir-weighted node-trace on the* S^2 *quotient.* The Phase 4B track α-D finding [18] is that

$$B(n_{\max}) = \sum_{n=1}^{n_{\max}} \sum_{l=0}^{n-1} (2l+1)\, l(l+1) \tag{10}$$

is the trace of the orbital-angular-momentum Casimir $\hat{L}^2$, weighted by the fiber dimension $2l + 1$, summed over the sectors of G^{S^2}. In lattice-gauge language, B is a *matter operator expectation value on the* S^2 *base*: the photon propagates on the sectors (the (n, l) labels), and the Casimir $\hat{L}^2$ is a gauge-invariant operator diagonal in this basis. $B = 42$ at $n_{\max} = 3$ is the value of this operator summed over the base nodes, weighted by their $U(1)$ fiber dimensions.

$F = \pi^2/6$*: Dirichlet zeta of the infinite Fock-degeneracy lattice.* The Phase 4F Track α-J result [18] is that

$$F = \sum_{n=1}^{\infty} \frac{n^2}{n^4} = \zeta_R(2) = \frac{\pi^2}{6}. \tag{11}$$

The weight $g_n = n^2$ is the Fock degeneracy of the n-th S^3 shell; the exponent $s = 4$ is the packing exponent $d_{\max} = \dim(\mathbb{R}^4)$ from Paper 0. This is not a finite-graph spectrum: it is the analytic continuation of a graph spectral sum to the full infinite Hopf lattice. In gauge-theoretic language, F is a *regularized gauge-field partition function* on the infinite base: a spectral zeta of the photon-sector degrees of freedom summed over all shells.

$\Delta = 1/40$*: Dirac-mode boundary count.* The Phase 4H Track SM-D result [18] is that

$$\Delta^{-1} = g_3^{\text{Dirac}}(S^3) = 2(n+1)(n+2)\Big|_{n=3} = 40. \tag{12}$$

This is the single-chirality Dirac mode count of the third Camporesi–Higuchi eigenspace on the unit S^3. In lattice-gauge-coupled-to-matter language, Δ is a *boundary-mode correction* counting the charged-spinor states at the truncation edge. Its entry with a minus sign in the combination $K = \pi(B + F - \Delta)$ is consistent (under the Paper 2 reading) with a vacuum-polarization correction that reduces, rather than enhances, the effective coupling at the cutoff.

B. What the framework observation claims

Observation 2 (Framework observation)**.** *Under the proposed Paper 2 identification of* $S^1 \to S^3 \to S^2$ *with (gauge phase)–(electron states)–(photon momenta), the Paper 2 combination rule* $K = \pi(B + F - \Delta)$ *is an additive combination of (i) a finite matter-operator trace on the* S^2 *base, (ii) an infinite gauge-field spectral zeta on the Fock lattice, and (iii) a boundary-mode count on the charged-spinor spectrum. All three quantities are intrinsic spectral invariants of the discrete lattice-gauge structure of the Hopf graph.*

Observation 3 (What the framework observation does *not* claim)**.** *The framework observation does not derive the additive combination rule itself. It does not explain why the three invariants combine with a common prefactor* π*, nor why* F *enters with a plus sign and* Δ *with a minus. The combination rule remains an empirical numerical match at* 8.8×10^{-8}*, with p-value* 5.2×10^{-9} *against an exhaustive combinatorial search (Paper 2) [18], and its status as an Observation is unchanged.*

C. Circulant structure reread

Paper 2 shows that the cubic self-consistency equation $\alpha^3 - K\alpha + 1 = 0$ is the characteristic polynomial of a traceless $\mathbb{Z}_3$-symmetric circulant matrix [18]. The $\mathbb{Z}_3$ action permutes three abstract coordinates; in Paper 2's interpretation, these are (B, F, Δ) or the three Hopf-bundle components. The $U(1)$ gauge structure forces Hermiticity of the circulant, which in turn forces the specific product relation $bc = K/3$ between off-diagonal entries.

In the present framework, the $\mathbb{Z}_3$ symmetry is the cyclic permutation of the three distinct *tiers* identified in Section IV.A: a finite matter trace, an infinite gauge zeta, and a boundary Dirac mode count. The Hermiticity arises from the compatibility of the $U(1)$ gauge structure across the three tiers. This is the same observation as Paper 2, rewritten in lattice-gauge language. It does not sharpen the observation; it does label its components with established gauge-theoretic names.

V. WHAT THIS PREDICTS, AND WHAT IT DOES NOT

A. What this does *not* produce

This paper does *not*:

- Derive the fine-structure constant α from first principles. Paper 2 is, and remains, an Observation with a proposed interpretation—not a derivation. The present paper shifts that content into gauge-theoretic language without removing any of its undetermined inputs.
- Prove the combination rule $K = \pi(B + F - \Delta)$. Phases 4B–4I plus two follow-on sprints eliminated twelve specific derivation mechanisms; the additive rule remains a numerical coincidence at $p = 5.2 \times 10^{-9}$ [18].

- Produce a continuum limit in which the edge Laplacian L_1 of the GeoVac graph converges to the Hodge–1 Laplacian on the unit S^3. Such a convergence is a mathematically well-posed but computationally nontrivial question; at present, we have only the finite-$n_{\max}$ edge-Laplacian spectrum (Section III.D).

- Give the photon dynamical content. The edge Laplacian is a combinatorial object; no dispersion relation, Lorentz structure, or QED vertex emerges from it alone.

B. What the framework observation does make available

What the observation does provide is a set of *natural next computations* that become well posed in the lattice-gauge frame but would have been ad hoc before:

Continuum-limit edge-Laplacian convergence. If the GeoVac graph node Laplacian L_0 converges to $-\Delta_{S^3}$ in the continuum (Paper 7), does L_1 converge to the Hodge–1 Laplacian on S^3? The continuous Hodge–1 spectrum on S^3 is known exactly: eigenvalues $n(n+2)$ with degeneracy $2n(n+2)$, $n \geq 1$, and zero β_1 (Ikeda–Taniguchi [11]). The finite-$n_{\max}$ graph has $\beta_1 = 2$ at $n_{\max} = 3$, $\beta_1 =$? at higher $n_{\max}$ (monotone or asymptotic behavior computable). If $\beta_1/N_1 \to 0$ and the nonzero spectrum converges to $n(n+2)$, we have a discretization of the Hodge–1 structure on S^3 compatible with the Hodge–0 convergence of Paper 7.

Higher Hopf bundles. The S^5 Bargmann–Segal lattice of Paper 24 [24] is built on the Hardy space $\mathcal{H}^2(S^5)$ of the holomorphic SU(3) $(N, 0)$ representations. $S^5 \to \mathbb{CP}^2$ is the next complex Hopf fibration after $S^3 \to S^2 = \mathbb{CP}^1$. Does the Bargmann–Segal graph carry an analogous SU(3) non-abelian gauge structure? Candidate check: the SU(3) Casimir values on the $(N, 0)$ irreps, weighted by fiber dimensions and summed under an analogous selection principle, produce a non-abelian analog of $B = 42$. Paper 24's HO rigidity theorem [24] already rules out calibration-π content on the S^5 graph, consistent with the expectation that a non-abelian SU(3) gauge-like structure would give a structurally different (likely pure-rational) combination rule, rather than an α-like transcendental.

Rank-2 Breit extension. Paper 22's Sprint 2 extension to the rank-2 Breit interaction opens a second family of edges carrying rank-2 tensor content [23]. In the Hodge vocabulary, this is a 2-cochain structure built on top of the 1-cochain $U(1)$ connection. A natural question: does the combined 1-form + 2-form simplicial structure carry a higher-form gauge theory (e.g., a 2-form B-field of the kind familiar in string compactifications)? This is speculative; we flag it only to point out that the lattice-gauge frame opens a coherent question where previously there was just an observation about integral sparsity.

Paper 2 tests independent of α. If the Paper 2 interpretation holds, the $U(1)$-Hopf structure of the GeoVac graph should show up in observables beyond α. Candidates within the GeoVac pipeline: (i) systematic relations between the spin-orbit splittings computed in Tier 2 (Paper 14 Sec. V [20]) and the edge-Laplacian structure of the spinor-bundle graph of Paper 18 Sec. IV [21]; (ii) the rank-2 Breit tensor contribution to ERI density (Paper 22 Sprint 2 extension [23]) interpreted as a 2-form contribution to the gauge-sector Hamiltonian. These are testable against the existing GeoVac codebase and give α-independent pressure on the Paper 2 interpretation. We leave the tests to future sprints.

VI. RELATED WORK

A. Lattice gauge theory on curved manifolds

Wilson [2] formulated lattice gauge theory on flat hypercubic lattices in Euclidean $\mathbb{R}^4$, with matter on sites and gauge fields on links. Luscher [3] developed the topology of lattice gauge fields on discretized S^4, with continuity constraints ensuring a well-defined integer topological charge; Berg and Luscher [4] used S^4 lattices for instanton studies. These are *compactification* strategies, not claims that S^3 or S^4 is a *natural* geometry for the physics. Aharony *et al.* [5] formulated large-N Yang–Mills on $S^3 \times \mathbb{R}$ as a probe of the Hagedorn/deconfinement phase transition, again as a finite-volume/infrared-regulator choice rather than as a natural arena. The category distinction matters: these works treat the manifold as a regulator; the GeoVac framework treats it as the physical arena of the electron bound states (via Fock's projection [1]), *and* as a geometric regulator of the gauge degrees of freedom.

Witten [6] formulated Chern–Simons theory on S^3 and computed the Jones polynomial of knots from its partition function. This is a *topological* gauge theory in three spacetime dimensions without local dynamical degrees of freedom. The GeoVac lattice-gauge reading would be a dynamical 3+1-dimensional theory (matter + photon propagation); the shared vocabulary of "connections on S^3" is suggestive but not structurally identical.

Recent work (post-2015) on simplicial lattice gauge theory (e.g., work on tensor-network formulations and categorical symmetries) has developed abstract combinatorial gauge theories on simplicial complexes, but without the specific physical motivation (Fock projection) that the GeoVac graph carries.

B. Graph Hodge theory

Lim [12] and Schaub *et al.* [13] articulated the combinatorial Hodge theory of graphs and simplicial complexes in modern form, motivated by applications in signal processing, ranking, and random walks. The mathematical framework is standard (going back to Eckmann in the 1940s and to algebraic topology generally), but its packaging in a form usable by engineers and data scientists is recent. The Hodge Laplacian $L_1 = B^\top B + B_2 B_2^\top$, its kernel as β_1, and the exact/coexact/harmonic decomposition of edge functions are all standard content of these surveys. The gap is that these surveys do not connect to lattice gauge theory or to quantum field theory on graphs; the communities are disjoint.

C. Conformal Maxwell theory and the AdS/CFT boundary

The conformal invariance of source-free Maxwell theory in 4D is a classical result of Cunningham [8] and Bateman [9]. Under conformal compactification of Minkowski space, the conformal boundary is $S^3 \times \mathbb{R}$, which is also the conformal boundary of AdS_5 (Maldacena [7]). Eastwood and Singer [10] developed a conformally invariant Maxwell gauge on this boundary; the photon modes expand in vector spherical harmonics on S^3. The continuous Hodge–1 spectrum on S^3 (Ikeda–Taniguchi [11]) is the relevant continuum limit.

Under the Paper 2 interpretation, the GeoVac S^3 (from Fock's momentum-space projection of the electron) and the conformal-compactification S^3 (from the conformal boundary of AdS_5) are *distinct* manifolds that happen to carry the same topology. Whether they can be unified into a single mathematical object—so that one S^3 carries both matter and gauge content in a coherent way—is an open foundational question; it is not resolved by the present observation. The present observation is more modest: the GeoVac graph, as a finite combinatorial object, carries both matter and gauge content simultaneously via its node-Laplacian / edge-Laplacian structure, regardless of how the continuum limit is ultimately interpreted.

D. Why the gap remained

Three communities would have had to overlap for the framework observation to be routine:

1. *Lattice gauge theorists*, who know Wilson's formulation but work on flat hypercubic lattices, not on Fock-projected momentum-space S^3 lattices.

2. *Graph Hodge theorists*, who know L_0/L_1 but work on engineering applications (ranking, signal processing), not on physical lattice gauge theory.

3. *Atomic/quantum chemistry theorists*, who know Fock's 1935 projection but treat it as a solved analytical curiosity, not as a computational framework underlying a discrete gauge theory.

The GeoVac project arrived at (3) for purely computational reasons (sparse qubit Hamiltonians from angular-momentum selection rules; see Paper 14 [20]). The overlap with (1) and (2) was invisible from inside any single community. Recognizing it from within the GeoVac pipeline required the Paper 2 observation, which in turn required the Phase 4B–4H structural decomposition to identify the three spectral homes of B, F, Δ. We flag the gap to place future GeoVac work at the intersection deliberately, rather than leaving the connection implicit.

VII. OPEN QUESTIONS

A. Gauge structure on the Bargmann–Segal S^5 lattice: answered (mixed)

Paper 24 [24] constructs the Bargmann–Segal lattice of the 3D isotropic harmonic oscillator on the Hardy space $\mathcal{H}^2(S^5)$ of SU(3) $(N, 0)$ symmetric representations. Its nodes are labeled (N, l, m_l), its edges are the $\Delta N = \pm 1$, $\Delta l = \pm 1$ dipole transitions. The underlying bundle is $S^5 \to \mathbb{CP}^2$, the next complex Hopf fibration. The question posed in the first draft of this paper was whether the Bargmann–Segal graph carries an analogous gauge structure, with three candidate outcomes: (a) abelian $U(1)$ analog, (b) non-abelian SU(3) analog, or (c) no gauge structure. Sprint 5 Track S5 (April 2026) resolved the question with a *mixed* verdict: (a) is confirmed, (b) is not natural, and (c) holds on the spectral side (the m_l-quotient is not a $\mathbb{CP}^2$ discretization). The data sharpen Paper 24's Coulomb/HO asymmetry rather than bridge it. Paper 24 Corollary 4 (the gauge-structure-reduction corollary added by the same sprint) documents the result in the Paper 24 vocabulary; we summarize here in the Paper 25 vocabulary.

Positive transfer: the abelian $U(1)$ Wilson–Hodge structure is universal. At $N_{\max} = 5$, the Bargmann graph has 56 nodes and 165 edges; the signed incidence matrix B, node Laplacian $L_0 = BB^\top$, and edge Laplacian $L_1 = B^\top B$ are well-defined; the first Betti number $\beta_1 = E - V + c = 110$ counts independent cycle classes (graph counts at $N_{\max} = 2$–5 and the plaquette censuses pinned live in `tests/test_su3_wilson_s5.py`, class `TestPaper25S5Pins`). Under a node-local $U(1)$ gauge transformation $\psi_v \to e^{i\chi_v}\psi_v$, the dipole edge amplitudes transform covariantly, exactly as in Section III.C for the S^3 case. The combinatorial Hodge decomposition and the SVD identity between nonzero spectra of L_0 and L_1 apply unchanged. In short: the combinatorial Wilson–Hodge vocabulary of Section II generalizes verbatim to the S^5 Bargmann graph (and, more generally, to any finite triangle-free simplicial 1-complex with a chosen orientation).

Negative: no natural non-abelian SU(3) *lattice structure.* A Wilson lattice gauge theory requires each link variable to live in a fixed group acting on a fixed matter irrep. The Bargmann graph has *different* SU(3) irreps on different shells—the dimension of $(N,0)$ grows as $(N+1)(N+2)/2$—and the dipole operator intertwines $(N,0)\otimes(1,0)\to(N+1,0)$ as a Clebsch–Gordan coupling, not as an SU(3) group element. The natural gauge group of the Bargmann graph is therefore $U(1)$, not SU(3). An SU(3)-covariant reformulation (replacing (N,l,m_l) labels by full weight-space labels of $(N,0)$ and promoting edges to SU(3) link variables) would be a different construction, and would not correspond to a Wilson lattice on the S^5 Hardy space in any natural sense.

Negative: the m_l-quotient is not a $\mathbb{CP}^2$ discretization. Collapsing the m_l-fibers at fixed (N,l) yields a 12-sector graph at $N_{\max}=5$ with sectors $\{(0,0),(1,1),(2,0),(2,2),(3,1),(3,3),(4,0),(4,2),(4,4),(5,1),(5,3),(5,5)\}$ and fiber dimensions $(2l+1)=1,3,1,5,3,7,1,5,9,3,7,11$. Its Laplacian eigenvalues are $\{0,2.22,4.87,4.94,11.65,12.11,18.74,28.09,33.58,50.61,63.77,99.43\}$ (12 distinct values, multiplicity scheme). The scalar Laplace–Beltrami spectrum on $\mathbb{CP}^2$ with the Fubini–Study Kähler metric of constant holomorphic sectional curvature $K=4$ is $\lambda_k=4k(k+2)$ with degeneracy $(k+1)^3$, giving $\{0,12,32,60,96,140,\ldots\}$ with multiplicities $\{1,8,27,64,125,216,\ldots\}$. A least-squares fit of the eleven nonzero quotient eigenvalues to the $\mathbb{CP}^2$ sequence under a single linear rescaling leaves a 40.8% maximum relative residual (measured against the fitted values); the ratios $\lambda_{\mathrm{quot},k}/\lambda_{\mathbb{CP}^2,k}$ range between 0.082 and 0.185, so *no* single rescaling can bring the maximum relative residual below the sharp two-point floor $(r_{\max}-r_{\min})/(r_{\max}+r_{\min})\approx 38\%$ under either normalization (a fortiori ≥33% against the data; deviation *factor* $\sqrt{r_{\max}/r_{\min}}\approx 1.50$) — there is no uniform fit under any convention. (An earlier version quoted a 24.98% residual from an unrecorded fit convention, and a first corrected printing overstated the against-the-fit floor as ≈50% — the one-sided minimax is the 38% two-point floor above; both are superseded by the pinned computation in `tests/test_su3_wilson_s5.py`, class `TestPaper25CP2Quotient`, which also pins the twelve-sector spectrum above.) The quotient is a combinatorial SO(3)-labeled graph, not a spectral discretization of the Kähler $\mathbb{CP}^2$.

Structural interpretation: two-layer Coulomb/HO asymmetry becomes three-layer. Paper 24 established the Coulomb/HO asymmetry at the level of the *spectrum* (on S^3, $L_0=D-A$ yields the Coulomb spectrum in the continuum limit via the $\kappa=-1/16$ projection; on S^5, L_0 is spectrally inert and the HO spectrum is in the diagonal). The present sprint adds a third layer at the level of *gauge-structure content*: the abelian $U(1)$ Wilson vocabulary is universal, but its physical content is not. On S^3, the dictionary (matter propagator, photon propagator) $=(L_0,L_1)$ is nontrivial—both sides carry physical spectrum and physical gauge content. On S^5, the dictionary is trivial on the matter side (the HO spectrum is not in L_0) and only combinatorial on the gauge side (L_1 generates cycle classes, but there is no "photon propagator on S^5" interpretation tied to the HO physics).

An immediate consequence is that the Paper 2 combination rule $K=\pi(B+F-\Delta)$ has no natural S^5 analog. A candidate Fock-degeneracy Dirichlet analog $F_{S^5}=\sum_{N\geq 1} g_N^{\mathrm{HO}}/N^{d_{\max}}$ with $g_N^{\mathrm{HO}}=(N+1)(N+2)/2$ and packing exponent $d_{\max}=\dim\mathbb{C}^3=6$ evaluates to $\frac{1}{2}[\zeta_R(4)+3\zeta_R(5)+2\zeta_R(6)]$, a mixed even/odd-zeta combination with no clean $\zeta_R(2)=\pi^2/6$ projection. The analogs of B and Δ are spectrum-computing-tied (both rely on the Coulomb projection), which Paper 23's Fock projection rigidity theorem restricts to S^3 alone. Together with Phase 4H's SM-running negative result [18] for alternative origins of Δ, the combined picture is that the S^5 graph is a lattice gauge structure with abelian $U(1)$ content but without α-like spectral content. This is consistent with Paper 24's operator-order transcendental discriminant: first-order complex-analytic operators give linear spectra, linear projections, and π-free lattices; second-order Riemannian operators give quadratic spectra, nonlinear projections, and π from spectral zeta regularization. Only the second-order Coulomb case carries Wilson-structure content of the α-relevant kind.

What remains open. The sprint answers the three candidate outcomes listed above; one narrow follow-up remains, but is well outside the Paper 25 frame: *if* the Bargmann graph were rebuilt in a genuinely SU(3)-covariant basis (not the SO(3) adaptation of Paper 24), with nodes labeled by SU(3) weight-space coordinates and edges carrying SU(3) intertwiners rather than $U(1)$ phases, *would* a different gauge-structure dictionary emerge? The answer is technically yes (this is classical representation theory), but the construction is no longer a Wilson lattice in the natural sense, and it has no known tie to the harmonic oscillator's physical spectrum. We record this as a speculative direction for future work and do not pursue it here. Data backing the Sprint 5 verdict: `debug/s5_bargmann_segal_graph.py`, `debug/s5_edge_laplacian_analysis.py`, `debug/data/s5_graph_spectrum.json`, `debug/s5_gauge_structure_memo.md`.

Sprint ST-SU3 update: pure-gauge SU(3) Wilson on Bargmann is well-defined; matter coupling rediscovers the CG obstruction. Sprint ST-SU3 (May 2026, `debug/st_su3_wilson_memo.md`, with module `geovac/su3_wilson_s5.py` and tests `tests/test_su3_wilson_s5.py` 49/49 pass, two slow-marked) revisits the non-abelian question with a structurally different construction: link variables $U_e\in\mathrm{SU}(3)$ as external 3×3 matrices in the *fundamental* representation, independent of the shell index N. This is categorically distinct from Sprint 5's adjacency-preserving SU(3) action on the $(N,0)$ tower (which the negative above ruled out): Wilson links do not require an action on shell labels. The construction is well-defined on the Bargmann graph as a Wilson lattice gauge theory: link variables in SU(3), plaquette holonomies in SU(3), action gauge-invariant under node-local SU(3) (verified to machine precision at $N_{\max}=2,3$), Haar-normalizable partition function. The maximal-torus reduction yields a coupled $U(1) \times U(1)$ Wilson action — the rank-2 sibling of the present paper's $U(1)$ on S^3 — with character $(1/3)\big(\cos a_P + \cos b_P + \cos(a_P + b_P)\big)$ verified at machine precision. The weak-coupling kinetic coefficient is $1/12 = 1/(4N_c)$ per plaquette per generator component, the $N_c = 3$ specialization of Paper 30's universal SU(N) formula. *However*, coupling Wilson SU(3) to matter on the Bargmann shells (in the natural $(N, 0)$ irreps) requires a CG intertwiner $\mathbf{3} \otimes (N, 0) \to (N+1, 0)$ — exactly the Sprint 5 Track S5 obstruction reappearing on the matter-coupling side. Wilson SU(3) on Bargmann is therefore a self-consistent *pure-gauge* theory; coupling it to natural matter requires the same CG structure that the strict Wilson formalism does not accommodate.

The structural reading that emerges from combining Sprint 5, Paper 30, and Sprint ST-SU3 is that Paper 24's Coulomb/HO asymmetry sharpens from three layers to *four*: the Wilson–Hodge combinatorial vocabulary is universal (transfers to S^5); the spectrum-computing role of L_0, the calibration π, and Wilson gauge *with natural matter coupling* are all Coulomb-specific. The previous version of § VII A (Paper 24 Corollary 4 & the three-layer reading) has Wilson gauge and matter coupling bundled together; the present update splits them: Wilson gauge alone is universal; only Wilson gauge with natural matter coupling on the shell tower is Coulomb-specific. The corresponding update to Paper 24 § V is documented at the same memo (§ 5.4).

B. Rank-2 Breit tensor: a 2-form gauge structure?

Paper 22's Sprint 2 extension treats the rank-2 Breit interaction as a genuinely new tensor sector of the GeoVac graph, with sparsity density depending on $l_{\max}$ but structurally distinct from the rank-0 Coulomb sector [23]. In the present framework, the rank-0 Coulomb edges are 1-cochains (a $U(1)$ connection on nodes); the rank-2 Breit edges would be 2-cochains (a 2-form on faces or pairs of edges).

A 2-cochain sector on the GeoVac graph would correspond (in the continuum) to a B-field of the kind familiar from string compactifications, or to the 2-form part of a higher-form gauge theory. Its combinatorial Laplacian is $L_2 = B_2^\top B_2 + B_3 B_3^\top$, if triangle and tetrahedral faces are included. Whether the Breit tensor structure closes into a well-defined higher-form gauge theory on the GeoVac graph—or whether it is better viewed as a nonlinear coupling on the $U(1)$ sector—is an open question. The Sprint 2 computational data (ERI densities at rank-2 $l_{\max}$ values) are the input to such a test; the test itself has not been performed.

C. Observables beyond α

The Paper 2 interpretation, if correct, should have implications beyond the fine-structure constant. Candidate observables to test:

- *Magnetic moment anomalies.* The $U(1)$ Hopf structure places the g-factor in a specific combinatorial context (via the spinor-bundle analog of the $U(1)$ matter-gauge coupling). Paper 14 Tier 2 [20] computes relativistic spin-orbit splittings; whether these decompose into Hopf-base and Hopf-fiber contributions with the same $\mathbb{Z}_3$ circulant structure as Paper 2 is computable.

- *α-dependence at finite $n_{\max}$.* If $K = \pi(B + F - \Delta)$ is indeed the lattice-gauge invariant producing α^{-1}, then at $n_{\max} \neq 3$ the truncated $B(n_{\max})$ and $\Delta(n_{\max})$ differ and give different "α" values. The selection principle $B/N = 3$ uniquely at $n_{\max} = 3$ says that only this cutoff reproduces the physical α. This is a prediction: no other $n_{\max}$ produces a recognizable physical coupling. Paper 2's negative result on higher Hopf bundles is consistent with this selection criterion; a systematic $n_{\max}$ scan is the obvious next test.

- *Sharpness of the combination rule.* Under what class of modifications to G (e.g., edge reweightings, adding or removing specific transitions) does K remain within 10^{-7} of α^{-1}? A combinatorial robustness check would distinguish a genuine spectral invariant (stable under small perturbations) from a fine-tuned numerical coincidence (collapsing under any perturbation).

None of these tests has been performed. The present paper's contribution is to identify them as well-posed within the framework, not to execute them.

VIII. CONCLUSION

The GeoVac framework builds a discrete graph on the unit S^3 whose node Laplacian converges to the Laplace–Beltrami operator on S^3 in the continuum limit. Section III of this paper made the observation—previously implicit in Paper 2 but never stated as its own claim—that the same graph, together with its signed incidence matrix B, carries all the combinatorial ingredients of a lattice gauge theory. The node Laplacian $L_0 = BB^\top$ is the matter propagator; the edge Laplacian $L_1 = B^\top B$ is the discrete Hodge–1 Laplacian; the Hopf quotient collapsing m_l-fibers produces the S^2 base graph on which gauge-neutral propagation lives.

Under the Paper 2 proposed interpretation, the three components of the Hopf bundle acquire physical labels: S^3 as electron bound states, S^1 as $U(1)$ gauge phase, S^2 as photon momentum sphere. The Paper 2 combination rule $K = \pi(B + F - \Delta)$ becomes a cross-sector spectral invariant of the lattice-gauge structure: B is a Casimir-weighted matter trace on the S^2 base, F is a spectral zeta of the photon-sector modes in the infinite-shell limit, and

Δ is a Dirac-mode boundary count. The additive rule itself remains an Observation—observed, not derived—just as in Paper 2; what changes is the vocabulary available to talk about it.

What is sharp in this paper is the mathematics: the GeoVac graph *is* a discrete Hodge 1-complex, *is* a Wilson lattice gauge structure under the natural $U(1)$ edge-phase assignment, and *has* an exactly computable S^2 quotient that is Paper 2's base graph. What remains a proposal is the physical interpretation of the bundle components and of the additive combination rule. The separation is deliberate and honest: the mathematical decomposition needs no further justification, while the physical reading remains Paper 2's proposed interpretation.

The framework observation does not resolve the open problem of why $K = \pi(B + F - \Delta)$ reproduces α^{-1}. It does locate the three quantities in a single mathematical context, and it does point to natural next questions: continuum-limit convergence of the edge Laplacian, non-abelian generalizations on the Bargmann–Segal S^5 lattice, higher-form analogs via the Breit rank-2 tensor, and independent observables beyond α. These are the tractable research directions that the observational synthesis makes well-posed.

The broader takeaway is that the GeoVac project arrived at lattice gauge structure without intending to: the requirements of a sparse, π-free, quantum-number-labeled graph for computational quantum chemistry happened to coincide, at the Hopf base of S^3, with the incidence structure that lattice gauge theorists know as a Wilson lattice with a $U(1)$ connection. That the same graph carries both content is either a coincidence of independent constructions or a signal that the physical arena of Coulomb bound-state quantum mechanics is, in its discrete form, exactly a lattice gauge theory of matter coupled to $U(1)$ gauge fields on the Hopf base. The present paper does not adjudicate between these two readings. It documents that both are consistent with the established mathematics, and it points to the follow-up computations that could distinguish them.

ACKNOWLEDGMENTS

This paper was developed using an AI-augmented agentic workflow, with implementation and documentation drafting performed collaboratively with Anthropic's Claude. The framework observation benefited from the Phase 4B–4I alpha-sprint series, which systematically identified the three spectral homes of the (B, F, Δ) ingredients over approximately nine months of sustained computational decomposition.

[1] V. Fock, Z. Phys. **98**, 145 (1935).
[2] K. G. Wilson, Phys. Rev. D **10**, 2445 (1974).
[3] M. Lüscher, Commun. Math. Phys. **85**, 39 (1982).
[4] B. Berg and M. Lüscher, Nucl. Phys. B **190**, 412 (1981).
[5] O. Aharony, J. Marsano, S. Minwalla, K. Papadodimas, and M. Van Raamsdonk, Adv. Theor. Math. Phys. **8**, 603 (2004).
[6] E. Witten, Commun. Math. Phys. **121**, 351 (1989).
[7] J. M. Maldacena, Adv. Theor. Math. Phys. **2**, 231 (1998).
[8] E. Cunningham, Proc. London Math. Soc. **8**, 77 (1909).
[9] H. Bateman, Proc. London Math. Soc. **8**, 223 (1910).
[10] M. Eastwood and M. Singer, Phys. Lett. A **107**, 73 (1985).
[11] A. Ikeda and Y. Taniguchi, Osaka J. Math. **15**, 515 (1978).
[12] L.-H. Lim, SIAM Review **62**, 685 (2020).
[13] M. T. Schaub, A. R. Benson, P. Horn, G. Lippner, and A. Jadbabaie, SIAM Review **62**, 353 (2020).
[14] M. Marcolli and W. D. van Suijlekom, "Gauge networks in noncommutative geometry," J. Geom. Phys. **75**, 71 (2014), arXiv:1301.3480.
[15] C. I. Perez-Sanchez, "Bratteli networks and the Spectral Action on quivers," arXiv:2401.03705 (2024).
[16] C. I. Perez-Sanchez, "Comment on 'Gauge networks in noncommutative geometry'," arXiv:2508.17338 (2025).
[17] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2025).
[18] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," GeoVac Paper 2 (2025).
[19] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2025).
[20] J. Loutey, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," GeoVac Paper 14 (2026).
[21] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).
[22] J. Loutey, "The geometric vacuum: Quantum structure from compact manifold topology," GeoVac Paper 21 (2026).
[23] J. Loutey, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," GeoVac Paper 22 (2026).
[24] J. Loutey, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," GeoVac Paper 24 (2026).
[25] GeoVac Track Q1-B, "Edge Laplacian of the GeoVac S^3 graph at $n_{\max} = 3$," internal memo, April 2026.
[26] GeoVac Project, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," Paper 28 (2026).
[27] GeoVac Project, "SU(2) Wilson Lattice Gauge Structure on the GeoVac Hopf Graph: The Non-Abelian Sibling of Paper 25," Paper 30 (2026).